

Children's Narrative Participation in Story Co-Creation with AI vs. Human Partners: Response Styles and Narrative Features

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Abstract

In this article, we examine response styles and narrative features among 73 children aged 4–8 during oral story co-creation with either an AI agent or a human partner. Children in both conditions generally communicated directly, but those with the AI partner were more likely to give brief responses, while those with a human partner provided more detailed and extended replies. Furthermore, within each response style, both shared and distinct narrative features emerged depending on the storytelling partner. These findings provide insights into how children engage with AI versus human partners in story co-creation and offer key design considerations for developing AI agents that adapt to children's diverse response styles, fostering richer, more engaging storytelling interactions.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Social and professional topics** → **Children**.

Keywords

Child-AI interaction, Conversational AI, Large Language Models, Interactive storytelling, Response styles, Narrative features

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1 Introduction

Narrative competence is widely recognized as a foundational skill for children that supports children's academic achievement and social development [20, 21]. Storytelling plays a crucial role in

nurturing this competence, catalyzing children's language development [2, 13], and facilitating imagination and sense-making [3]. Traditionally, children do storytelling in the classroom with peers and teachers or at home with siblings and guardians. Advances in artificial intelligence (AI) have expanded the pool of potential storytelling partners for children, providing opportunities for personalized support beyond what is traditional parental involvement [5, 22, 28]. While prior studies [28] highlight the efficiency and comparable learning outcomes of AI agents in supporting different learning activities (e.g., storytelling), less is known about how children engage with these agents, particularly in terms of their response styles and the depth of their narrative contributions.

This work-in-progress paper explores children's narrative participation while co-creating stories with an AI agent compared to a human partner, including their response styles and narrative features. Drawing on 73 story co-creation transcripts from children aged 3 to 8, we conducted both quantitative and qualitative analysis to understand similarities and differences in children's storytelling behavior with AI and human counterparts. Our work answers the following questions:

- **RQ1:** What types of responses styles do children use when co-creating a story with AI vs. human partners? Do children's response styles differ across contexts?
- **RQ2:** What are the similarities and differences in the narrative features within each response style used by children when co-creating a story with AI vs. human partners?

2 Related Works

2.1 The Role of Storytelling

Storytelling is a highly engaging activity that facilitates children's language development, communicational skills and listening comprehension [9, 10, 14–16, 18]. Compared to activities like storybook reading, storytelling relies on spoken words, structured narratives, and mental imagery, requiring active engagement from both the storyteller and the audience [19]. In an experimental study, Isbell et al. [12] found that children in the storytelling group demonstrated stronger story comprehension during retellings, while those in the story reading group exhibited more complex language use. Similarly, Trostle et al. [23] showed that children who engaged in

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storytelling scored higher on vocabulary and comprehension than those who only listened to stories. Despite these benefits, access to rich storytelling experiences is often limited by constraints such as reduced instructional time, high student–teacher ratios, and the limited availability of caregivers at home.

2.2 Storytelling with Conversational Agents

Advancement in AI, particularly with GPT-4, have transformed the way children engage with storytelling technologies [15]. Recent studies [4, 6, 27, 28] show that AI-powered conversational platforms can support children’s cognitive development through interactive storytelling experience. Compared to earlier systems that relied on predefined scripts, current large language model (LLM)-powered systems [7, 17, 28] generate more natural, context-sensitive interactions, allowing for a more engaging and personalized storytelling process. These systems can pose open-ended questions, respond in nuanced ways, and adapt the storyline in real time based on children’s input. While these technologies show promise in empowering children, it remains critical to understand how children’s interactions with AI compare to those with traditional human storytelling partners.

Prior research has found important differences in how children communicate with human versus agents. Aeschlimann et al. [1] and Gampe et al. [11] found that children share less information with voice assistants than they do with humans. Similarly, Xu et al. [1, 26] found that while children spoke more clearly when interacting with a conversational agent, those reading with a human adult used more varied vocabulary, gave longer responses, and stayed more on topic. Although both groups performed similarly in response accuracy, the gap in verbal engagement widened with cognitively demanding questions. While storytelling is widely recognized as a powerful tool for language and narrative development, little is known about how children verbally engage during story co-creation with different types of partners.

3 Method

3.1 Participants

A total of 73 children participated in the study, with 40 assigned to the AI condition and 33 to the human condition. Participants ranged from 4 to 8 years old ($M = 6.61$, $SD = 1.31$), and 54.8% were female. The sample was predominantly White (71.2%), with others identifying as mixed race (20.6%), Asian (4.1%), Black (1.4%), or other/declined to state (2.7%). Additionally, 6.9% identified as Hispanic or Latino. No significant differences were found between the AI and human conditions across demographics, including age, gender, race/ethnicity, parental education, or AI use history. Full demographic details by condition are in Appendix A.

3.2 Story Co-creation Session

Each session lasted up to 30 minutes and followed a consistent structure across both conditions (human and AI, see Figure 1). It began with a story setting phase, where the storytelling partner prompted the child to provide key elements such as the main characters, their name, and the location. The partner initiated a storyline, which then progressed through turn-taking: the partner asked a question (e.g., about the character(s)’ next action), acknowledged

the child’s response, and incorporated it into the unfolding narrative. To ensure consistency, both the AI and human partners were instructed to ask similar open-ended main and follow-up questions about the main character(s)’ next steps and what happens next in the storyline.

In the human condition, an experimenter sat beside the child and co-created the story in real time. The experimenter was a trained research assistant with no prior relationship to the child participants, ensuring that the children’s engagement in the story co-creation process was not influenced by prior familiarity. In the AI condition, the child interacted with a smart speaker. Children were also encouraged to choose a plush toy and act out the story. All sessions were conducted in English, videotaped, and audio was recorded and transcribed for analysis.

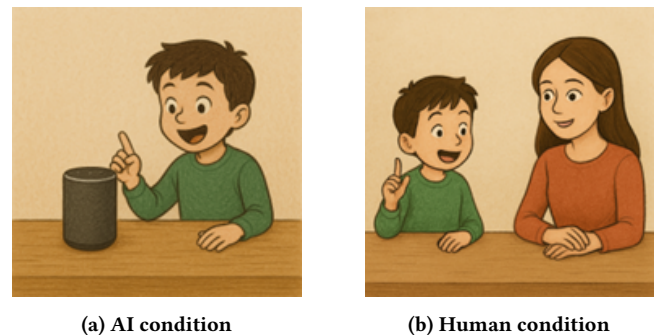


Figure 1: Story co-creation in AI and Human conditions (ChatGPT, Mar 28, 2025, prompted with “A scenario where a child tells stories with Alexa; a scenario where a child tells stories with a human”).

4 Data Analysis

To examine children’s response styles during story co-creation sessions, the first author conducted inductive open coding of children’s responses, first identifying whether a child’s response directly addressed the question and then characterizing them as brief, detailed, or extended. The research team met weekly to iteratively review, clarify, and refine the codebook, ensuring a shared understanding. Once the inter-rater reliability reached 89.5%, the finalized codebook (see Appendix B) was used for independent coding. After coding children’s response styles, two authors conducted a follow-up analysis to examine the narrative features within each style. Each independently reviewed the associated responses and wrote analytic memos capturing their observations on children’s narrative complexity and depth. The authors then met to compare memos and synthesized recurring narrative features, including single action, sequential actions, contextualization, and justification (e.g., using language such as ‘because,’ ‘so,’ or ‘just in case’). Using this refined set of categories, they independently applied codes to the responses, organized by response style. Each author reviewed the other’s work, provided feedback, and resolved discrepancies through weekly meetings.

5 Preliminary Findings

5.1 Children's Response Styles with AI vs. Humans

We examined the distribution of children's response styles when interacting with AI versus a human partner (see Figure 2). Notably, in both conditions, most children's response styles are either direct answers or no response/"I don't know", with relatively fewer indirect answers or other styles.

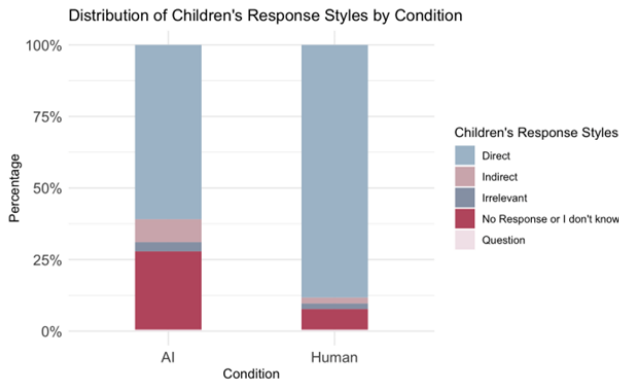


Figure 2: Distribution of Children's Response Styles by Condition.

Most subtypes of direct response styles varied significantly between AI and human interactions (see Table 1). Specifically, children were significantly more likely to provide brief responses when interacting with AI than with a human partner ($p < 0.05$). In contrast, they were significantly less likely to give detailed responses ($p < 0.001$) or responses with extensions ($p < 0.001$) when interacting with AI.

Table 1: Number of Children's Direct Response Subtypes by Condition

Direct Response Subtypes	AI	Human	Chi ²
Brief response	188 (40.3%)	183 (48.9%)	$p < 0.05$
Detailed response	73 (15.7%)	98 (26.1%)	$p < 0.001$
Extended response	23 (4.9%)	50 (13.3%)	$p < 0.001$
Total	284	331	

5.2 Characteristic Narrative Features of Response Styles

Given that most children's response styles are direct when interacting with either AI or a human storyteller, we further analyzed these direct responses, particularly in terms of the narrative features within these direct responses in both conditions. We noticed that children's responses in both conditions predominantly used goal-oriented actions, meaning that regardless of the depth of their responses (brief, detailed, or extended), children mainly focused on what the main character will do next to advance the story. This

is likely due to the nature of the storytellers' prompt, which were action-based questions (e.g., "What do you think Lily will do next to find the legendary tree?"). However, within those goal-oriented actions responses, we observed different narrative features across different types of response styles, conditions and age groups. In the following sections, we will compare and contrast the narrative features within each response style to highlight the similarities and differences in children's narratives, revealing varying levels of depth and complexity that should be considered when designing future AI tools.

5.2.1 Narrative Features within Direct and Brief Response. The narrative features within goal-oriented actions are similar when children's responses are direct and brief, regardless of whether the storyteller is AI or human. In both cases, the responses mention only one action without much detail. For example, when the AI asked a four-year-old kid, "What do you think Hawa will do next to fulfill her special mission?", the child's answer was, "go outside into the forest." Similarly, when the human storyteller asked a child with the same age, "What should Own do if he is only half as strong as the wall?", the child responded, "He should get another hammer." These instances exemplify children's focus on a single, straightforward action in their goal-oriented responses, illustrating a recurring pattern of direct and brief response style with their story partner.

5.2.2 Narrative Features within Direct and Detailed Response. The narrative features in children's direct and detailed responses varied depending on whether they were co-creating a story with an AI or a human storyteller. In the AI condition, children tended to respond either with a sequence of actions alone or with a combination of actions and descriptive details. These response patterns emerged consistently across different age groups. By contrast, most children in the human condition offered responses that intertwined actions with descriptive information about those actions.

Approximately 51% followed a pattern of sequential actions, in which children listed two or more steps without additional elaboration or contextual detail. For example, when AI asked what Uni and Lucy would add to their exploration list, a five-year-old replied, "Going to the library, and then we'll go to the Costco," presenting a simple sequence of destinations. In comparison, the remaining 49% of responses featured a mix of other narrative features. Many of these included contextualization, where children added setting details or descriptive information about objects or conditions in addition to the main actions. For example, a child said, "I would say get a strong, strong sheet of the blocker made out of metal," providing specific material details that help the listener understand what kind of blocker is being described. A smaller portion of responses involved anticipated consequences or reasoning, in which children not only stated what the character(s) would do but also explained what might happen as a result or offered a rationale for the action.

However, in the human condition, both younger and older children tended to provide more descriptive information in their responses, along with sequential actions, rather than simply offering straightforward actions without much elaboration. 65% of older children's responses and 63% of younger children's responses included additional context or descriptive detail. For example, when a child who was asked, "What do you think they're going to see in the dark forest?" The child responded, "I think they are gonna see

and find a werewolf like in the forbidden forest in Harry Potter.” This response illustrates the child’s ability to incorporate elements from familiar stories into their own narrative, which enriches the setting and adds specificity to what the main characters might encounter next.

5.2.3 Narrative Features within Direct and Extended Response. Distinct narrative patterns emerged in children’s direct and extended responses when co-creating stories with either an AI or a human partner. In the AI condition, 81% of children’s responses across age groups were goal-oriented with justification, meaning they included reasons, anticipated consequences, or stated purposes behind characters’ actions. Interestingly, all younger children in the human condition produced responses that mirrored the dominant pattern seen in the AI condition. To provide a concrete example of this pattern, we turn to a four-year-old who collaborated with the AI during a battle preparation scene. The child proposed: “*They would need a lot of knights so they could fight with a lot of swords, and they could fight the knight down, and the dragon down, the dragon, and all the dragon come to, so all dragons came.*” This response illustrates how the child justified the need for “a lot of knights” by reasoning about the upcoming battle and anticipating the arrival of multiple dragons.

In contrast to the detailed justification-focused responses observed in the AI condition and among younger children in the human condition, 55% of older children’s responses in the human condition exhibited the most sophisticated narratives in the dataset. These responses featured a broader range of narrative elements, including imaginative elaboration, dialogue and creative problem-solving. In these 55% of responses, older children actively expanded the storyline, introduced multiple events, and incorporated vivid descriptions to shape the plot. Their contributions reflected a higher level of narrative sophistication and creative engagement. For example, one 8-year-old child described how the main characters would reach their destination in a highly imaginative and structured way:

First, they use a magic spell to take off their mermaid legs for only ten minutes and replace them with normal legs and make Broccoli a ginormous dinosaur. That can make him a ginormous dinosaur. And the girl, the elf Tadril, rides on it, and they go and secure the gem. But they only have a little bit time to get back to the real land before their swimming ability. So they get to the back of the shore, do the magic spell again, and poof, they’re in the water, and they’re swimming really fast, but then the water is out. They don’t have time. So they survived to the surface. Broccoli put the gem. Broccoli ate the gem because it was shaped like broccoli. And then when he got on land, he had a broccoli clung out of his, and he ate it. And it was his dinner—it was his meal for the rest of his life.

This child introduced a creative solution involving a magic spell that temporarily transforms the characters’ mermaid legs and turns Broccoli into a ginormous dinosaur. This imaginative plot enabled the characters to ride the dinosaur to retrieve the gem. Then, the child added an obstacle “a 10-minute time limit” to create urgency and drove the plot forward. As the story unfolded, the child ended with a humorous twist: “Broccoli ultimately eats the gem because

it is shaped like broccoli.” Notably, this child incorporated fantasy elements, unexpected obstacles and a humorous ending. This narrative demonstrates the child’s higher level of creative thinking and competence to construct a complex and imaginative narrative.

Similarly, another seven-year-old child responded to a question “Where do you think Joanna and Floral go in order to find the magical rainbow with money?” by offering a creative and well-sequenced storyline:

First they went to the cotton candy forest, which were trees, but, like, cotton candy. So they searched and searched. Then Joanna knew they would need a guide, so Flora went to call her best friend, but she wasn’t home. So Flora used her magic powers, and then she told Joanna that the horn would go brighter when they got closer to the magical.

The child began their story by setting the scene with the imaginative “cotton candy forest,” where the trees resembled cotton candy. The plot unfolded as the characters search for a solution, facing an obstacle when Joanna realized that they need a guide. Flora initially tried to use a realistic approach by calling her best friend but turns to magic when that failed. The story concluded with Flora guiding Joanna by observing the horn’s brightness. The child incorporated these key story elements: a setting, goal, obstacle, and creative solution, while balancing creativity and logic, demonstrated through the playful metaphor and fantasy elements.

6 Discussion and Future Work

This study explores how children aged 4–8 engage in oral story co-creation with either an AI agent or a human partner, focusing on their response styles and narrative features. Our findings reveal significant differences shaped by the storytelling partner and the children’s developmental stage.

Children interacting with the AI agent were more likely to give brief, uncertain, or indirect responses, while with human partners, they provided more detailed and extended responses. This finding aligns with previous experiments that show children have more substantial and direct responses when answering human partners’ questions during dialogic reading compared to AI questions [24, 26]. We also observed that children’s narrative complexity varied depending on their response style. Brief responses, which were similar across conditions, consisted of simple actions without elaboration. However, children’s responses in the AI condition varied, with some focusing on sequential actions and others on descriptive details, without a clear preference for one single approach. In contrast, with human partners, most children consistently provided one or two actions with descriptive elaboration. This difference may stem from children recognizing that human partners, being physically present, are more likely to engage with additional details, prompting children to elaborate and ensure clearer understanding of their ideas [25].

The most complex narrative features emerged in extended responses. Younger children and those interacting with AI agents tended to offer justifications (e.g., reasons or consequences), while older children with human partners developed more sophisticated narratives, introducing settings, obstacles, and solutions. Older children offered the richest answers, aligned with their developmental

stage and cognitive skills [8]. However, older children in the AI condition did not exhibit the same level of narrative complexity. This may be explained by their perception of AI as a less responsive partner, which could limit the engagement.

Identifying these narrative features helps us understand how children create meaning in storytelling and how their interactions with AI or human partners shape the quantity and complexity of their responses. Our findings highlight significant differences in verbal engagement, especially in response length, depth, and elaboration, underscoring the gap between child–AI and child–human communication. These differences call for AI–powered storytelling tools that foster richer, context-aware interactions. Future research should focus on discourse analysis to deepen our understanding and inform the design of more effective AI tools for children.

7 Selection and Participation of Children

Our study received approval from the Institutional Review Board of the University of Michigan. Participants were children recruited from two public libraries in a mid-western U.S. city between July and January 2024. A total of 73 children took part in the study, which involved story co-creation activities with either an AI or a human partner. All participants provided consent to participate and were informed that they could withdraw at any time. Upon completion, children who finished the study received a storybook, while their parents were given a \$60 Visa gift card as compensation for their participation.

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A Demographic Background of Participants

Table 2: Background Information by Study Conditions

	Overall Sample	AI	Human	ANOVA/Chi2
<i>Gender</i>				$\chi^2(2) = 2.45, p = 0.29$
Female	54.8%	47.5%	63.6%	
Declined	1.4%	2.5%	0.0%	
<i>Age</i>	6.61(1.31)	6.46(1.28)	6.78(1.33)	$F(1, 71) = 1.06, p = 0.31$
<i>Race</i>				$\chi^2(4) = 4.53, p = 0.34$
White	71.2%	75.0%	66.7%	
Mixed	20.6%	15.0%	27.3%	
Asian	4.1%	2.5%	6.1%	
Black	1.4%	2.5%	0.0%	
Other/Declined	2.7%	5.0%	0.0%	
<i>Ethnicity</i>				$\chi^2(1) = 0.06, p = 0.81$
Hispanic/Latino	6.9%	7.5%	6.1%	
Non-Hispanic/Latino	93.2%	92.5%	93.9%	
<i>Parental Education</i>				$\chi^2(3) = 2.64, p = 0.45$
Above Bachelor's	48.6%	50.0%	46.9%	
Bachelor's	36.1%	40.0%	31.3%	
Below Bachelor's	13.9%	10.0%	18.8%	
Declined	1.4%	0.0%	3.1%	
<i>Use of AI Agents</i>				$\chi^2(5) = 1.80, p = 0.88$
Daily	29.2%	32.5%	25.0%	
Weekly	25.0%	22.5%	28.1%	
Monthly	11.1%	10.0%	12.5%	
Less than monthly	9.7%	7.5%	12.5%	
Yearly	8.3%	7.5%	9.4%	
Never	16.7%	20.0%	12.5%	
<i>N</i>	73	40	33	

B Codebook

Table 3: Codebook for Children’s Response Styles

Code	Descriptions	Examples
DAE (Direct Answer, Extended)	The child provides a direct and relevant response to the question posed by the AI or human partner and includes additional information that goes beyond what was required or expected, thereby extending the conversation.	Q: What do you need to climb over the wall? A: A ladder because then you can get up a tree, and then they can look where the mountain is. The cave is then jumped on the cave. Then they can go down the cave, and then enter the cave.
DAD (Direct Answer, Detailed)	The child provides a direct and relevant response to the question posed by the AI or human partner, enriched with detailed information. These responses often include multiple descriptive elements, clarifications, or step-by-step reasoning that deepen the original answer.	Q: What equipment do you think they need? A: They need a touch sword with a strong handle and wooden texture. Q: What can Love and Malika do together? A: They could go on a magical adventure, share their favorite things or create a secret handshake. Your turn. A: Play together and make things and tell stories.
DAB (Direct Answer, Brief)	The child provides a direct and relevant response to the question posed by the AI or human partner. The response is concise, typically consisting of a single word or short phrase, without elaboration or additional detail.	Q: What equipment do you think they need? A: They will need to find a sword.
IA (Indirect Answer)	The child provides an indirect response that is related to the general topic but is not directly or loosely connected with the expected answer.	Q: What kind of story do you think Love will add to their collection next? A: They can draw together, play with some stories, and go camping.
Irrelevant Response (IR)	The child provides a response that is off-topic or unrelated to the question.	Q: What would mountains and volcanoes look like then? A: Just ocean. So cold it would squeeze the, so the carbon is frozen, and if it tries to do it, it would rock it, bam.
NA	The child does not provide a response or respond <i>"I don't know"</i> .	—